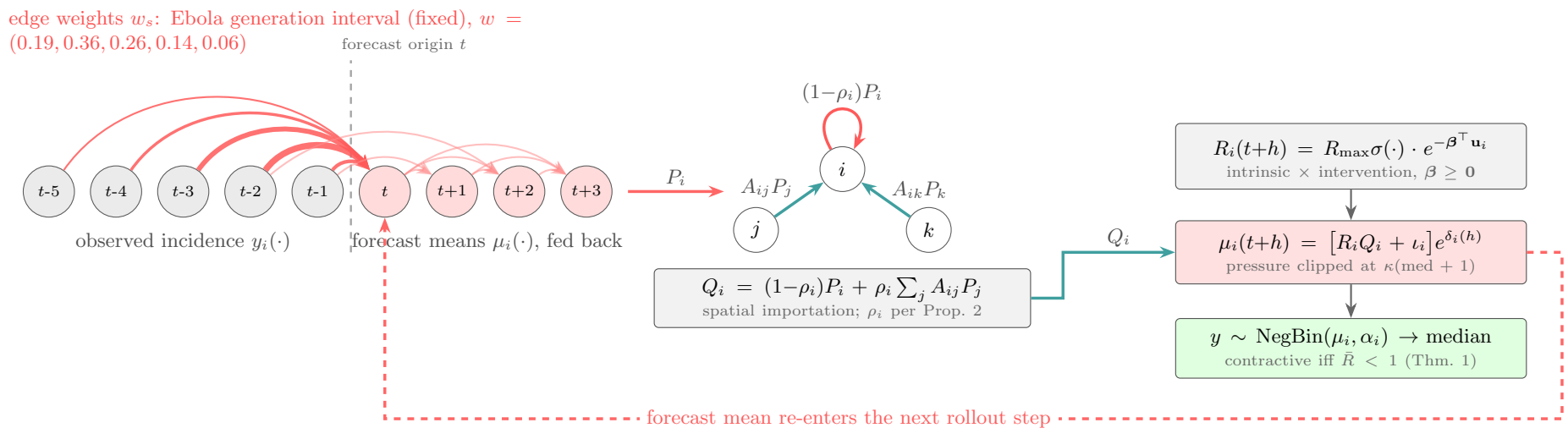

edge weights w_s: Ebola generation interval (fixed), $w = (0.19, 0.36, 0.26, 0.14, 0.06)$
forecast origin t
t-5
t-4
t-3
t-2
t-1
t
t+1
t+2
t+3
observed incidence $y_i(\cdot)$
forecast means $\mu_i(\cdot)$, fed back
P_i
$(1-\rho_i)P_i$
i
$A_{ij}P_j$
$A_{ik}P_k$
j
k
$Q_i = (1-\rho_i)P_i + \rho_i \sum_j A_{ij}P_j$
spatial importation; ρ_i per Prop. 2
Q_i
$R_i(t+h) = R_{\max}\sigma(\cdot) \cdot e^{-\boldsymbol{\beta}^\top \mathbf{u}_i}$
intrinsic × intervention, $\boldsymbol{\beta} \geq \mathbf{0}$
$\mu_i(t+h) = \left[R_i Q_i + \iota_i\right] e^{\delta_i(h)}$
pressure clipped at κ(med + 1)
$y \sim \text{NegBin}(\mu_i, \alpha_i) \rightarrow \text{median}$
contractive iff $\bar{R} < 1$ (Thm. 1)
forecast mean re-enters the next rollout step